Enclosure

The enclosure plays an extremely important role in the success of the PPS units. As shown in Fig. 3, the enclosures of the most well-known brands are made of high-quality plastic, which seems to be slightly against logic, as heat removal is best done in a metallic enclosure. However, modern plastics can provide good enough heat removal capabilities (UL94 V-0 type plastics), while ensuring that the user experience is great as far as look and feel is concerned.

As in all other products, enclosure design is an integral part of the product design. In addition to important facia details, such as location and size of the indication and control apparatus, location of the input and output ports and many minute details—such as location of the carrying handle, path for the air passage, location of forced cooling fans, presence of an ambient light source—have huge impact on the overall customer experience. As energy capacity of the PPS grows, the enclosure needs to incorporate metallic parts and even drawing wheels to ease the portability of the units.

An important aspect of enclosures would be to ensure that there is enough free space left for a clear air circulation path. This space could be calculated from the worst case power dissipation values (in watts), which would give an idea of the temperature rise that may happen in locked stationary air and then deducing the amount of air to be displaced per unit time, which will lead to sizing of the free air space and the fans used for forced cooling.

Indication and control

Most modern day systems have some kind of a screen to show its status and to allow the user to make control selections. Although there are not much control selections that the user can make in a PPS, it has been a convention in almost all PPS units to have an LCD or a TFT screen which provides detailed status information about the unit. The typical items shown on the screens are:

1. Battery charge indication in percentage of the overall Wh capacity
2. Battery charging and discharging indication
3. Time of complete discharge to show how long the PPS unit's battery will last at the present discharge current levels
4. Time to full charge to show how long it would take for the PPS battery to be fully charged at the present charging current level
5. Charging input levels for each type of charging input
6. Discharging source indication and level of discharge for each

The typical control selections are starting and stopping of different discharge sources, such as AC and DC output switches.

Power conversion

The power conversion components form the heart of the entire product. As shown in Fig. 4, the power conversion architecture includes the charging controller, battery and BMS, AC to DC rectifiers, DC to AC inverter, and DC to DC converters at both input and output ports.

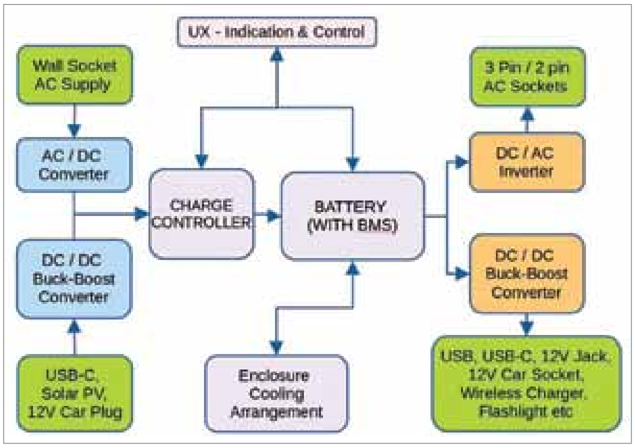

Fig. 4: A block diagram view of a typical portable power station

There are multiple product teardown videos online, and most of these videos show that the PPS units have purpose-built PCBs, which provide the most efficiency and integration of all capabilities that the PPS offers. This often means that the designers would need to create PCBs which could be used in multiple products, with specific hooks to select specific capabilities based on the product model. This integration also aids in the maintenance and repairs of PPS units and reduces e-waste as there are fewer PCBs to recycle.

Some PPS units also allow multiple units to be chained (chainable units) with other similar units to provide longer back-up or higher power. This capability should be easy to implement as long as the BMS in both the units can talk to each other over side-band connections provided in special chaining cables. Such a chaining would put the batteries inside the PPS units in parallel.

While prevalent in the discharging mode, we are yet to come across a chainable PPS that allows chained charging of multiple units from a single source. This may be because charging times are reduced when they are charged individually.

There are numerous possibilities in this space. Beyond the civilian use cases, PPS units may be of help in defence and paramilitary activities as these activities require active use of electrical power at places where the electricity grid may not be available. EFY